

Problem 3

Given a set of observations sequence,
how do we learn the model probabilities that
would generate them?

$$O = \{O_1, O_2, \dots, O_T\} \quad \lambda = \{A, B, \pi\} ?$$

Reminder

We have the following tools:

$$\alpha_t(i) = P(O_1, O_2, \dots, O_t, q_t = s_i \mid \lambda)$$

$$\beta_t(i) = P(O_{t+1}, O_{t+2}, \dots, O_T \mid q_t = s_i, \lambda)$$

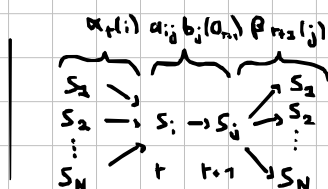
$$\gamma_t(i) = P(q_t = s_i \mid O, \lambda)$$

Baum-Welch algorithm

Searches for a local optima of $\bar{\lambda}$.

We need a new tool:

$$\begin{aligned} \xi_t(i, j) &= P(q_t = s_i, q_{t+1} = s_j \mid O, \lambda) \\ &= \frac{\alpha_t(i) a_{ij} b_j(O_{t+1}) \beta_{t+1}(j)}{P(O \mid \lambda)} \\ &= \frac{\alpha_t(i) a_{ij} b_j(O_{t+1}) \beta_{t+1}(j)}{\sum_i \sum_j \alpha_t(i) a_{ij} b_j(O_{t+1}) \beta_{t+1}(j)} \end{aligned}$$



From which we can derive: $\gamma_t(i) = \sum_j \xi_t(i, j)$

Also deriving following:

- expected number of times over all t that s_i was visited: $\sum_t \gamma_t(i)$
- expected number of times over all t we transitioned from s_i to s_j : $\sum_t \xi_t(i, j)$

How do we use these to improve $\bar{\lambda}$?

At each iteration we can estimate from observations and parameters:

$$\begin{aligned} \bar{\pi}_i &= \gamma_1(i) & \bar{a}_{ij} &= \frac{\sum_t \xi_t(i, j)}{\sum_t \gamma_t(i)} & \bar{b}_i(v_k) &= \frac{\sum_{s.t. O_t = v_k} \gamma_t(i)}{\sum_t \gamma_t(i)} \end{aligned}$$

→ Given $\lambda = \{A, B, \pi\}$ and O we can produce $\alpha_t(i), \beta_t(i), \gamma_t(i), \xi_t(i, j)$
Given $\alpha_t(i), \beta_t(i), \gamma_t(i), \xi_t(i, j)$ we can produce $\bar{\lambda} = \{\bar{A}, \bar{B}, \bar{\pi}\}$ →